MIL-B-36488
7 June 1966

MILITARY SPECIFICATION

BOOTH, AUDIOMETRIC EXAMINATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one size and type of prefabricated audiometric examining booth, with left or right hand door, suitable for use in the testing, calibration and recording of aural acuity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

HH-I-563	Insulation, Mineral-Wool; Blanket, Felt, and Industrial-Batt (for Heated Surfaces).
QQ-S-698	Steel, Sheet and Strip, Low Carbon.
QQ-S-775	Steel, Sheets, Carbon, Zinc-Coated.
TT-C-598	Compound, Calking; Plastic (for Masonry and Other Structures).
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall.

Military

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center .

20531 Booth, Audiometric Examination

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

UNDERWRITERS' LABORATORIES, INC.

Underwriters' Laboratories Standards

UL 57 Electric Lighting Fixtures.
UL 62 Flexible Cord and Fixture Wire.
UL 498 Attachment Plugs and Receptacles.
UL 507 Electric Fans.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories Inc., Walt Whitman Road, Melville, L.I., N.Y. 11749; 207 E. Ohio Street, Chicago, Ill. 60611; and 1655 Scott Blvd., Santa Clara, Calif. 95050.).

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3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), prior to the start of production of the contract quantity, one preproduction sample of the audiometric examination booth shall be assembled, completely finished, and tested by the contractor. The cognizant Government inspection office shall be notified when the preproduction sample is available for testing to determine compliance with requirements specified herein. The performance tests shall be conducted and verified by the National Bureau of Standards, or by the Riverbank Acoustical Laboratories, Armour Research Foundations, Geneva, Ill., or another testing laboratory acceptable to the contracting officer. A detailed laboratory test report, indicating compliance with the specification, shall be submitted to the contracting officer for approval. Written approval of the materials, construction and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited. If the supplier has furnished an acceptable audiometric booth for the identical item within two years preceding the date of contract, and proposes to furnish booths identical with those previously accepted, then the requirement for submission of a preproduction sample may be waived upon request.

3.2 Fire and casualty hazard.

3.2.1 The contractor shall submit to the contracting agency prior to delivery, proof that the equipment he will supply under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. UL57, UL62, UL498 and UL507. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

3.2.2 In lieu of the label or listing, the contractor may submit prior to delivery, independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.2.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from

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complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.3 Materials. Materials not definitely specified shall be of the highest quality and suitable for the purpose intended. Material or material treatment which cause toxic reaction or produce unpleasant odor in the booth, shall not be used.

3.3.1 Carbon steel. The sheet and strip cold rolled carbon steel used in the fabrication of the walls and channel members shall conform to chemical composition 1009 of QQ-S-698.

3.3.2 Insulation. The insulation used in the construction of the booth shall be vermin, fire, and mildew resistant acoustical material conforming to HH-I-563.

3.3.3 Gaskets. The door gaskets shall be fabricated from resilient, adhesive back strips of sponge rubber.

3.3.4 Drag seal. The drag seal affixed to the bottom of the door, shall be fabricated of spring phosphor bronze.

3.3.5 Door hardware. The hinge and latch provided for use with the door, shall be fabricated of heavy duty forged brass, with a satin chrome finish.

3.3.6 Dessicant. The dessicant provided with the booth windows shall be of the chemical bagged type conforming to MIL-D-3464.

3.3.7 Calking compound. The calking compound used in the erection of the prefabricated booth shall conform to Grade 1 of TT-C-598.

3.4 Design. The prefabricated audiometric examining booth shall be designed to permit handling and storing of the unassembled components without special handling equipment in a minimum of storage space. The booth shall be capable of withstanding erection and disassembling not less than three times without adversely affecting the sound excluding capabilities. The assembly and disassembly of the booth shall be performed without the use of special tools and without damage to any of the seams which are used in assembling, except the seals made by the calking compound. The erected booth shall be capable of being transported by a fork lift truck without structural damage.

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3.4.1 Sound reduction qualities. The composite sheet metal and sound absorbing panels shall have a minimum over-all noise reduction coefficient (N.R.C.) of 0.95 and the following sound absorption coefficients:

Frequency cycles per second (cps)	Sound Absorption Coefficients
125	0.70
250	0.99
500	0.99
1000	0.99
2000	0.94
4000	0.83
N.R.C. (av.)	0.95

3.4.2 Sound pressure levels. The ambient sound pressure levels in the audiometric examining booth shall not exceed the following:

Octave Bands (cps)	150 300	300 600	600 1200	1200 2400	2400 4800	4800 9600
Sound Pressure Level decibels (db)	40	38	40	48	57	67

3.4.3 Noise reduction. The minimum allowable noise reduction of the modular components assembled as a complete booth and tested in a reverberant field shall be as follows: (plus or minus 3 db for instrument error).

Octave Bands	37.5	75	150	300	600	2100	2400	4800
CPS	75	150	300	600	1200	2400	4800	9600
Noise Reduction db	22	28	36	46	53	58	61	63

A separate noise reduction test on the blower itself shall be conducted. The noise level of the blower shall be at least 9 db less than the noise level in the room.

3.4.4 Radio frequency suppression. The audiometric examining booth shall provide effective radio frequency suppression throughout the entire spectrum. When tested as specified in 4.4.2, the booth shall provide an average attenuation of 52 db from 0.1 megacycles (mc) to 100 mc.

3.5 Construction. The construction of the audiometric examining booth shall be in strict accordance with DPSC Drawing 20531 and as specified herein. Sufficient acoustical insulating material (see 3.3.2), four inches wide and one inch thick, shall be supplied to fill all channels of all panels during erection. A calking gun and four tubes of calking compound (3.3.7) shall be provided for use in assembling the booth. A molded rubber floor mat shall be provided, of such dimensions, to fit snugly from wall to wall and yet be

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easily removed for cleaning. The floor mat shall be corrugated and not less than 3/16 inch thick. A radio frequency copper mesh screen, properly framed for attachment on the exterior of the window, shall be provided with each booth. One spray can of each color paint used on the booth shall be provided for the purpose of touching up scratches incurred during the erection process. All hardware required to complete the assembly of the booth, shall be supplied with each booth.

3.5.1 Wall panels. The wall panels shall consist of cold rolled steel sheet, on the inside and outside surfaces. The inner and outer panel walls shall be reinforced with not less than three equally spaced channel shaped stiffeners, spaced on 24-inch centers. One inch deep channels shall be located along all edges of the perimeter of the finished panel. The inner steel sheet, shall be 0.0299 inch thick (No. 22, United States Standard Revised gage (U.S.S. Rev. gage), and shall be perforated with 3/32-inch-diameter openings on 3/16-inch staggered centers. The outside surface, the reinforcement channels, and the channels along the perimeter of the panel shall be of 0.0598-inch thick (No. 16, U.S.S. Rev. gage) carbon steel. The wall panels shall be welded and riveted into a unit to form a rugged metal sheathed acoustical panel. Before the interior skin is attached, the panel shall be built up in a laminated construction consisting of the solid exterior steel skin, dampened with multiple sandwich construction, bonded and isolated with a viscoelastic material, and then filled with acoustical material (see 3.3.2). The multiple sandwich construction shall weight not more than 7 pounds per square foot nor less than 6 pounds per square foot. After the acoustical sandwich material has been bonded in place in the panel, all space between the steel skins, framing, and stiffeners shall be filled with the insulation which shall be cut slightly larger and thicker than the shape and space required to be filled so as to require forcing into location. The insulation shall be packed so as to leave no voids or empty pockets. The total thickness of the layer or layers of insulation before installation shall be in excess of the space to be filled, so as to require the insulation to be held in a state of compression in the completed structure. The compression shall be sufficient to hold the insulation in place without any other attachment and shall prevent movement or settling of the insulation under severe conditions of vibration and jarring encountered in handling, shipping, or in operation. A minimum number of insulation batts shall be used and shall be tightly fitted with all joints staggered.

3.5.1.1 Window. One side wall panel of the booth shall have a double glazed window of 1/4-inch thick clear safety glass, 24 inches high and 30 inches wide. Each pane shall be mounted in an acoustically tight one piece rubber molding containing a sealing insert which shall be securely mounted in the interior and exterior steel skins of the panel, The window shall be adequately framed with channel shaped members of 0.0598-inch thick (No. 16 U.S.S. Rev. gage) rolled steel. A bag of desiccant shall be made fast to the top of the window frame between the two panes of safety glass.

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3.5.2 Floor panel. The floor panel shall consist of a floor plate, an underside sheet, channel shaped reinforcing members, and shall be framed with the same material as the top plate. All members shall be cold rolled carbon steel. The floor plate, the reinforcing members and the channels along the perimeter shall be 0.1196-inch thick (No. 11, U.S.S. Rev. gage). The underside sheet shall be 0.0598-inch thick (No. 16, U.S.S. Rev. gage). The reinforcing members shall be welded to the underside of the floor plate to form a rigid structural floor. The floor panel shall be filled with acoustical material. Floor channels which accept the wall panels shall be premitered, prepunched on 6-inch staggered centers, calked, and plug welded to the perimeter of the floor plate assembly. The structural design of the floor shall not allow any perceptible deflection of the floor plate when two people (300 pounds minimum) occupy the booth.

3.5.3 Roof panel. The roof panel shall be constructed of the same steel sheet construction and acoustical materials as described herein for the wall panels, except that it shall be 6 inches thick. The roof panel shall be of the drop-in type and shall contain an integral built-in ventilation air discharge silencer which shall be acoustically compatible with the booth wall panels. There shall be no projections inside or outside the booth. The acoustic fill materials shall be fully contained behind perforated materials so that no particles can be carried into the air stream. The panel shall be capable of discharging the air volume as generated by the ventilating system (80 cubic feet per minute (CFM).

3.5.4 Door panel. One of the 48-inch wall panels shall contain a 4-inch thick soundproof door of the overlap type. The door shall be supplied with double acoustic seals around its entire periphery and shall be equipped with positive pressure hardware to assure tight seals. The same steel sheets and acoustical materials specified for the wall panels shall be used on the door. The inner sheet of the door leaf shall be fabricated of one-piece steel sheet with the periphery area solid and the inner face perforated. The door leaf shall be reinforced with three channel members spaced approximately 20 inches on centers joining the inner sheet of the door leaf to the outer sheet. Two additional channel stiffeners spaced approximately 6 inches from the top to the bottom of the door leaf and attached to the inside surface of the outer sheet shall be supplied. The three hinge connecting areas as well as the latch connecting area shall be reinforced with 0.1196-inch thick (No. 11,U.S.S. Rev. gage) plate for accepting 1/4-20 bolts to secure hardware to the door and frame assembly. Each booth door leaf shall have the hardware attached. The hardware for the booth door frame shall be supplied separately or as specified (see 6.1). The door shall be furnished with a refrigerator type door latch.

3.5.5 Door hardware. The door hardware shall consist of a positive pressure latch, complete with micrometer adjustments. The latch shall be designed and constructed to permit opening the booth door from the inside as well as the outside. Three hinges, containing case hardened, ball bearing rollers shall be provided with each door.

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3.5.6 Booth vibration isolation system. The entire booth shall rest on two properly loaded rubber-in shear-vibration isolator rails which shall extend from front to back on each side of the booth. The isolator rails shall be tack welded to the underside of the floor panel with 2-inch welds on 6-inch centers. The vibration isolator rails shall be designed for a natural frequency, under a 800-pound load, of 6-1/4 cycles per second (CPS). The rails shall consist of an upper 4-inch 5.4 pound structural steel channel, and a bottom plate 2-3/4 inches wide by 1/4-inch thick with rubber-in-shear isolator mounts spaced periodically along the length of the rails. The rails shall extend the full depth of the booth.

3.5.7 Connecting corner joints. Four connecting corner joint assemblies, constructed of 0.0598-inch thick (No. 16, U.S.S. Rev. gage) cold rolled steel sheet, shall be supplied. Each joint assembly shall be joined by spot welding on 2-inch centers. The design and fit of the corner joints which serve to connect the wall panels shall be such that they will prevent noise leakage into the room and shall maintain the attenuation level of the room itself. The corners shall be formed so that there is no direct passage for noise to go through the joint. Any noise must pass at least two 90-degree corners. The vertical corner joinst shall fit snugly along the entire height of the adjacent wall panels.

3.5.8 Ventilation system. A complete, wall-mounted, mechanical, forced-air, ventilation system, capable of supplying 80 CFM of air shall be supplied. The ventilation system shall operate below the binaural minimal audible field, 0 degrees azimuth threshold of hearing, when measurements of sound pressure levels are made 3 feet from the blower opening inside the booth. The ventilating system shall operate on a single phase, alternating current, power source of 110 volts or 220 volts, 50 to 60 cycles, as specified (see 6.1). The ventilation system shall be capable of circulating ambient air through the booth at the rate of 80 CFM with the door closed. The blower shall be provided with a power cord for plugging into a power supply receptacle located in the electrical box on the exterior of the light fixture wall panel. The ventilation system shall be mounted on the exterior of this panel through felt insulating strips with sheet metal screws at assembly.

3.5.9 Light fixture. The side wall panel opposite the window wall panel shall be provided with one swivel-type, 110-volt, medium-socket, incandescent light fixture. The fixture shall have two 110 volt convenience receptacles mounted on the base. The fixture shall be attached to a box in the panel on the inside of the booth. An exterior box shall contain a receptacle for attachment to an external 110 volt or 220 volt, as specified (see 6.1), alternating current power supply and a power outlet for the ventilation system. The light fixture shall be removable. Provision shall be made for firmly mounting the fixture 14 inches from the ceiling in the center of the wall panel opposite the window panel. The light fixture wall panel shall also be provided with a framed 5-inch-diameter opening to accommodate the ventilation system.

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3.5.10 Jack panel. A completely wired jack panel shall be provided beneath the window. The panel shall consist of two 1/8-inch thick black plastic plates, 6 phone-type jack receptacles with cover plates and an interconnecting device (see 6.6). The connecting wire shall be No. 18 Gage shielded wire.

3.6 Weight. The total weight of the floor, excluding vibration isolator rails, shall be 180 pounds ± 10 pounds. The weight of the roof panel shall be not less than 140 pounds or more than 160 pounds. The weight of the erected panels shall not exceed 1700 pounds nor be less than 1500 pounds. The weight of the wall-mounted ventilation unit shall not exceed 80 pounds. The weight of the completely assembled unit with all accessories shall not exceed 1800 pounds.

3.7 Joining. Unless otherwise specified herein or on the drawing, all sheet metal parts shall be joined by spot welding or riveting. Materials to be welded shall be clean and free of rust, scale, corrosion, oil, water, and any other matter injurious to welding. Spot weld shall be of good quality, free of cracks, porosity, tip pick-up, blown spots, and other defects which indicate that spot welds were made with dirty electrodes or improperly prepared surfaces. Individual spot welds shall be sufficiently fused to pull the spot out of the joint leaving a hole in one of the welded sheets (or strips). All scale or flux deposits shall be removed from the finished welds. Riveting shall be limited to joining the internal channel stiffeners to the perforated sheet where welding is not feasible.

3.8 Finish, metal surfaces.

3.8.1 Cleaning. All sheet metal surfaces subject to painting shall be prepared for painting including grinding, smoothing of welds, degreasing, and filling as needed. The sheet metal surfaces which face the acoustical fill shall have the oil film intact in order to protect this surface.

3.8.2 Prime coat. All exposed sheet metal surfaces, after cleaning, shall be primed with a nitrocellulose modified phenolic prime coating to a dry film thickness of 0.6 mil minimum.

3.8.3 Finish coat. All surfaces primed shall be thoroughly coated with a finish coat of nitrocellulose alkyd modified lacquer paint of the vendor's standard paint color. The exterior portion of the booth shall be painted a medium green, the interior light gray, and the ceiling, white.

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3.9 Identification marking. Each booth shall be permanently and legibly marked with the following information:

Manufacturer's name or registered trade mark.
Federal Stock Number
Contract or purchase order number.
The letters: U.S.

Markings secured by adhesives shall not be used.

3.10 Other marking. Since the sound pressure levels within the audiometric examination booth must not exceed the levels specified below, the vendor shall insert in the installation instructions a cautionary note warning the user that the booth must be located in sufficiently quiet area to attain the following maximum sound pressure levels within the booth:

Octave Bands	150	300	600	1200	2400	4800
CPS	300	600	1200	2400	4800	9600
Sound Pressure Level (db)	40	38	40	48	57	67

3.11 Service data. The contractor shall furnish, with each unit, two copies of service and assembly data containing complete instructions for installation, erection, operation, maintenance, a list of erection tools necessary, and a list of component parts. This information shall specifically include pictorial instruction for erection, drawings, electric power connections, and detailed instructions for installing the door for either left or right hand opening. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment (see 6.1). The preliminary service data shall be submitted sufficiently early to permit adequate review and to allow incorporation by the contractor of required revisions and corrections in the

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final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located. When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. If the supplier has furnished acceptable manuals for the identical item within two years preceding the date of contract, and proposes to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived (see 6.1).

3.12 Workmanship. All equipment furnished shall be clean, well made, and free from any defect which may affect appearance or serviceability.

3.13 Independent laboratory reports. Laboratory reports from a nationally recognized independent laboratory, adequately equipped and competent to perform such service (see 6.5), shall be supplied to the contracting agency as acceptable evidence that the item to be supplied has passed the requirements of this specification. Submission of such a report shall not absolve the contractor from complete compliance with requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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form a rigid structural floor. The floor panel shall be filled with acoustical

4.1.1 Inspection. Inspection, as used in this specification, is defined, as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	(AQL) (percent defective)
For visual examination		
Major defects	II	1.0
Total (major and minor defects combined) defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, for Inspection Level S-4. Acceptance number shall be zero regardless of sample size.

4.2.3 Raw materials. Four ounces of each lot of metal used in the construction of the booths shall be selected for chemical analysis.

4.3 Examination. The audiometric booths shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Any component part missing, bent, or otherwise unusable.
102 Any exposed edge or surface not free of burs or sharpness.
103 Any component does not fit properly with its mating member.
104 Any weld or rivet cracked, broken or not porperly cleaned.
105 Assembly instructions not complete, legible or otherwise not as specified.

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Minor

201 Any surface not free of scratches, roughness, or machine marks.
202 Identification marking not permanent, legible, or complete.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The prefabricated audiometric booths shall be examined for defects in dimensions.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Testing conditions. All tests shall be performed on a completely assembled booth located in a reverberant room. The noise level generated in the reverberant room, outside the test booth, shall be of such magnitude to fully realize the noise reduction specified herein without encountering instrumentation noise.

4.4.1.2 Testing methods. Octave bands of white noise shall be projected from two sets of loud-speakers, one in a lower corner of the reverberation room, and the other in the approximate center of an opposite wall and higher than the roof of the booth. Exploration with a sound level meter more than 5 feet from the loudspeakers shall show the variation in level from the mean throughout the reverberant room, including the space between the booth and the nearest wall, to be no more than 1/2 db. The nearest corner of the booth shall be more than 6 feet from a loudspeaker. Preliminary measurements shall be taken to show that microphone location is not critical. Exploration with a sound level meter within the booth and more than 1 foot from the inside walls shall indicate no variation greater than 1 db from the mean. Measurements shall be made with a single location of the booth microphone in line with the window, midway between the walls and 43 inches above the floor. Each octave band of sound shall be turned on for one minute and the sound pressure level in the room and in the booth, simultaneously and continuously recorded. After a complete run is made, the systems, including the microphone, shall be interchanged and a second run made. The results shall be averaged to minimize any differences in response between the two systems. Noise levels in the booth, with the signal turned off, shall be at least 10 db below those measured with the signal turned on.

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4.4.2 Radio frequency suppression test. The audiometric examining booth shall be tested to determine compliance with the radio frequency suppression requirements specified in 3.4.2. The booth shall be tested throughout the radio frequency (r.f.) spectrum using a spark-gap transmitter coupled to a whip antenna with counterpoise as the radiating element. The transmitter and radiating antenna shall be placed inside the booth and connected to a battery source to avoid extraneous radiation from incoming power lines. The receiving antenna shall be placed three feet from the radiating antenna and immediately in front of the door. Readings shall be taken of the ambient noise level with the transmitter off and door open, transmitter operating and door open, transmitter operating and door closed. Attenuation shall be determined by the following relation:

$$\text{ATTENUATION} = 20 \log \frac{E_1}{E_2}$$

where E_1 and E_2 are the field strength in microvolts, with transmitter operating and the door open and closed respectively.

5. PREPARATION FOR DELIVERY

5.1.1 Unit of issue. One booth, audiometric, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation and packing. Each unit shall be disassembled as follows:

5.1.2.1.1 Ventilation system. Ventilation system shall be removed from the room and packed in a box of appropriate size and design constructed in accordance with PPP-B-640. Box design shall include suitable interior fittings or shall contain sufficient cushioning material to prevent shifting of, or damage to, contents. Closure and strapping shall be in accordance with box specification.

5.1.2.1.2 Wall panels and door panel. The three wall panels and door panel shall be packed in an exterior container of appropriate size constructed in accordance with PPP-B-640, class 2, grade A. Design of box shall include a corner-cut tray securely fastened to a heavy-duty wood pallet. Box shall be fitted with two heavy wood end frames supported to edge panels of the unit.

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Point of contact between wood and panels shall be covered with grade A paper. The top of the box shall be reinforced with a channel fabricated of the same material as the box. Design shall further include an outer box made in half-slotted style (two pieces with overlapping end panels) securely fastened to wood pallet. Container design shall include suitable interior fittings or shall contain sufficient cushioning material to prevent shifting of, or damage to, the contents. Closure and strapping shall be in accordance with box specifications.

5.1.2.1.3 Roof and floor. The roof and floor shall be packed in an exterior container of the type shown for wall panels and doors.

5.1.2.1.4 Accessories and hardware. Accessories and hardware shall be packaged in a corrugated carton of appropriate size and design. Carton shall include interior fittings or shall contain suitable cushioning material to prevent shifting of, or damage to, the contents. Carton shall be placed within the exterior container containing the roof and floor in such a manner as to prevent movement.

5.1.2.1.5 Ventilation system. No further overpacking shall be required.

5.1.3 Level C.

5.1.3.1 Packaging and packing. The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Interior carton. Interior carton (hardware, etc.) shall be marked to identify contents.

5.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, set markings shall be applied in accordance with 5.3.2 and 5.5.2.1 of MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) The operating voltage and frequency required at using installation (see 3.5.8 and 3.5.9).
(c) Whether left or right hand door is required (see 3.5.4).
(d) If installation by manufacturer is desired.
(e) When preproduction sample is not required (see 3.1).
(f) Where service data is to be sent, when required (see 3.11).
(g) When submission of service data manuals for review and submission of 15 additional copies are not required (see 3.11).
(h) Selection of applicable levels of packing and packaging (see 5.1 and 5.2).

6.2 This specification covers the following item in the Federal Supply Catalog:

6515-065-9600 BOOTH, AUDIOMETRIC EXAMINATION

6.3 Definitions. For the purpose of this specification, noise reduction is defined as the difference between a sound pressure level in a reverberant room outside the booth and the sound pressure level inside the booth. The minimum allowable noise reduction of the modular components, completely assembled as a booth with the ventilating system installed, shall be as specified in 3.4.3 when measured in a reverberent room.

6.4 Independent laboratory tests equipment used. The noise is to be generated in a Grason-Stradler Noise Source, octave bands selected by means of an Allison Model 2BR variable band pass filter, amplified by a Western Electric power amplifier, and delivered to two sets of loud speakers. The resulting sound shall be detected with Western Electric 633-A microphones, and the signals delivered through identical Western Electric amplifiers, Daven-T-Pad attenuators and Allison Model 2BR variable band filters, to a Sound Apparatus Company Model RZ two-channel Graphic Level Recorder.

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6.5 Test reports. RF test compliance data in the form of a certified laboratory test report shall be obtained from an independent testing laboratory such as Interference Measurement Laboratory, Inc., Brooklyn, New York, or approved equal. The acoustic performance tests shall be conducted and verified by the National Bureau of Standards, or by the Riverbank Acoustical Laboratories, Amour Research Foundation, Geneva, Illinois, or approved equal.

6.6 Availability of components. Components listed in this specification are available from sources as follows:

Phone-type jacks - Switchcraft Part No. 12B-JAX
Cover plates for jacks - Switchcraft Part No. 515
Mounting plates - Franklin Fibre Lamitex Corp., Wilmington, Delaware Grade XX Lamitex
Interconnector - Cinch-Jones Part No. S303-RP and P303-RP
Interconnector - Cinch-Jones Part No. S304-RP and P304-RP

Custodians:
Army - MS
Navy - MD
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36488

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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